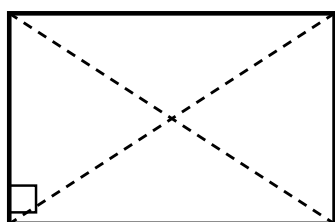


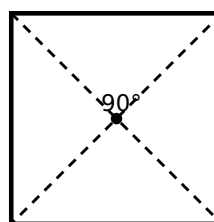
Rectangles & Squares

Both have four right angles. The key exam ideas are their diagonals — when they are equal, when they bisect each other, and when they meet at right angles.

Properties & diagonals



Rectangle



Square

Both diagonals are drawn dashed. In a square they also cross at 90° .

Rectangle

- All four angles are 90° ; opposite sides are equal and parallel.
- The two **diagonals are equal** in length.
- The diagonals **bisect each other** (cut each other in half).
- Diagonal length (sides l and b) = $\sqrt{l^2 + b^2}$.

Square

- All four sides equal **and** all four angles 90° (a square is a special rectangle).
- Diagonals are **equal, bisect each other**, and meet at 90° ; they also bisect the corner angles.
- Diagonal length (side a) = $a\sqrt{2}$.

Prerequisite refresher: "Bisect" means cut exactly in half. "Diagonal" joins two opposite corners.

Where marks slip away

Trap 1 — Rectangle diagonals aren't perpendicular. They are equal and bisect each other, but they only meet at 90° in a **square**.

Trap 2 — "Equal diagonals" setups. If two diagonal expressions are given, set them **equal** (rectangle diagonals are equal) and solve.

Trap 3 — Diagonal formula. Use $\sqrt{l^2 + b^2}$; don't just add $l + b$.

Trap 4 — Square is a rectangle and a rhombus. Every square is a rectangle (and a rhombus), but not every rectangle is a square.

Slip 5 — Leave $\sqrt{2}$ surd or round. $a\sqrt{2}$ is exact; if a number is wanted, use $\sqrt{2} \approx 1.41$.

Model answers — write them like this

Example A. The diagonals of a rectangle are $(2x + 1)$ and $(3x - 1)$. Find x and the diagonal length.

Step 1: In a rectangle the diagonals are equal: $2x + 1 = 3x - 1$.

Step 2: $1 + 1 = 3x - 2x$, so $x = 2$.

Step 3: Diagonal = $2(2) + 1 = 5$.

$\therefore x = 2$, diagonal = 5

Example B. Find the length of the diagonal of a square whose side is 7 cm.

Step 1: Diagonal of a square = side $\times \sqrt{2}$.

Step 2: $= 7\sqrt{2} \approx 7 \times 1.41 = 9.9$ cm.

$\therefore$ Diagonal = $7\sqrt{2} \approx 9.9$ cm

Example C. A rectangle has sides 6 cm and 8 cm. Find its diagonal.

Step 1: Diagonal = $\sqrt{l^2 + b^2} = \sqrt{6^2 + 8^2}$.

Step 2: $= \sqrt{36 + 64} = \sqrt{100} = 10$ cm.

$\therefore$ Diagonal = 10 cm

Next: [try the worksheet](#), then [check the answer key](#).